

54.1 and 2 - Homework

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January 2026

Homework 1.1

19. Do the three lines $x_1 - 4x_2 = 1$, $2x_1 - x_2 = -3$, and $-x_1 - 3x_2 = 4$ have a common point of intersection?

$$\left[\begin{array}{cc|c} 1 & -4 & 1 \\ 0 & 7 & -5 \\ 0 & -7 & 5 \end{array} \right]$$

- All three lines pass through this point
- Row 3 = -(row 2) $\implies$ linearly dependent equation
- Dependent equations are redundant, not free
- You drop row 3 because it doesn't restrict the solution set
- $x_2 = -\frac{5}{7}$, $x_1 = 1 + 4(-\frac{5}{7}) = -\frac{13}{7}$
- unique solution, consistent

Homework 1.2

9. Write down the equations corresponding to the augmented matrix and verify your answer is correct by substituting the solutions you obtained back into original questions

$$\left[\begin{array}{ccc|c} 0 & 1 & -6 & 5 \\ 1 & -2 & 7 & -4 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & -5 & 6 \\ 0 & 1 & -6 & 5 \end{array} \right]$$

- I didn't correctly recognize that x_3 is free, and in the general solution all we simply due is take the original equation and plug it back in. Thus getting the same answer
- $x_1 = x_2 - 6x_3 \rightarrow 5 + 6x_3 - 6x_3 = 5$ ■
- $x_2 = x_1 - 2x_2 - 7x_3 \rightarrow [6x + 5x_3] - 2[5 + 6x_3] + 7x_3 = -4$ ■

23. Choose h and k such that the system has a: no solution, b: unique, c: many solutions

$$\left[\begin{array}{cc|c} x_1 & hx_2 & 2 \\ 4x_1 & 8x_2 & k \end{array} \right]$$

- Brute force can't be a viable option IMO, need to go to OH
- Since its a simple 2×2 matrix, we can cross factor to get

$$\left[\begin{array}{cc} a & b \\ c & d \end{array} \right] \rightarrow ad - cb \rightarrow 8 - 4h = 0 \rightarrow h = 2$$

- By finding $h = 2$, We are determining exactly when no solution happens. The next step is to substitute 2 in the equation; then scale by the x_1 constant in the second linear equation i.e. in $R2$

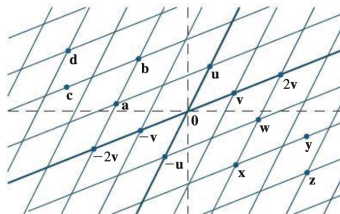
$$4(x_1 + 2x_2) \equiv R2 \text{ in the coefficient matrix}$$

- the above equation shows us that when $h = 2 \wedge k = 8 \implies$ Infinitely many solutions.
 1. Whats really cool is we just showed that they are on the same line, thus we can grab any number in the $\mathbb{R}$ for the cartesian plane
 2. This means if we chose $k \neq 8$ then they are a different line, no solution
- When $h = 2 \wedge k \neq 8 \implies$ No solution
- When $h \neq 2 \wedge \forall x \in \mathbb{N} \implies$ Unique solution

Homework 1.3

5. Simple question that demonstrates that x_1, x_2, x_3 times a vector A will give you the system of equations. Derived from vector equations
7. Find linear combination of a, b, c, d

Use the accompanying figure to write each vector listed in Exercises 7 and 8 as a linear combination of $\mathbf{u}$ and $\mathbf{v}$. Is every vector in $\mathbb{R}^2$ a linear combination of $\mathbf{u}$ and $\mathbf{v}$?



- Great description [here](#)
 - You can pick up u and v s.t. you plot the $\hat{u} \wedge \hat{v}$. Exactly how you plot for x and y cords.
 - Fractions are observed in this problem
 - Make sure to count the intersections of each line as one step!!
15. Whenever you see the $\text{Span}v_1, v_2$ questions, with language to list vectors and entries; it means you generate the answers of multiplying $x_1.x_2$ etc. Such a cool problem

$$v_1 = \begin{bmatrix} 7 \\ 1 \\ -6 \end{bmatrix} \quad v_2 = \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} \rightarrow$$

Scale the x_n to generate examples, adding them to get the linear combination; i.e.

$$(n)x_1 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} + (n)x_2 \begin{bmatrix} -5 \\ 3 \\ 0 \end{bmatrix} = [\text{Linear combination of entries x, y, z}]$$

17. For what values of h is b in the plane spanned by a_1, a_2
- Great walkthrough [here](#), essentially you solve with REF and add the scaled numbers to h, then solve for it if you have two cases of what it could be
29. Compute the center of gravity of the system consisting of the following point masses, where $m = m_1 + \dots + m_k \wedge \hat{v} = \frac{1}{m}[m_1v_1 + \dots + m_kv_k]$
- [Walkthrough](#)
 - We never learned dividing a system so I am confused if there is another way to approach this type of problem (i.e. when we add up the masses and divide it by the sum of the vectors)
 - ChatGPT explained: The center of mass formula is a weighted average of position vectors, where the masses act as scalar weights.
 - * Implying that you must scale each m_jv_j by its mass
 - * Add the resulting vectors i.e. $\sum m_jv_j$
 - * Divide the entire vector sum by total mass m by each entry in the vector
 - * Thus

$$\bar{v} = \frac{m_1v_1 + \dots + m_kv_k}{m}$$

$$\text{After adding all the scaled vectors} = \frac{17, -34, 16}{14} \rightarrow \begin{bmatrix} \frac{17}{14} \\ \frac{-34}{14} \\ \frac{16}{14} \end{bmatrix}$$

Homework 1.4

11. Given A and $\mathbf{b}$, write augmented matrix for linear system that corresponds to the matrix $A\mathbf{x} = \mathbf{b}$. Then solve the system and write the solution as a vector

– This problem is asking you to compute $\mathbf{x}$, where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

- The edge case is when you don't have a nice triangular system, and you must take parametric form and solve for remaining variables.
- For the edge case, we use variable t to imply that it lacks a pivot or when a linear equation is being re-arranged for RHS. I.e. $x_2 + 3x_3 = c \rightarrow x_2 = c - 3t$

17. How many rows of A contain a pivot position? Does the equation $A\mathbf{x} = \mathbf{b}$ have a solution for each $\mathbf{b}$ in $\mathbb{R}^4$

$$A = \begin{bmatrix} 1 & 3 & 0 & 3 \\ -1 & -1 & -1 & 1 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 4 & 1 & 2 \\ 0 & 1 & 3 & -4 \\ 0 & 2 & 6 & 7 \\ 2 & 9 & 5 & -7 \end{bmatrix}$$

19. Can each vector in $\mathbb{R}^4$ be written as a linear combination of the columns of the matrix A above? Do columns of A span $\mathbb{R}^4$

- You don't need to calculate each RREF $\forall x_n \in B$
- Rather, you solve REF for A and then determine if there is a pivot in every row. If true $\rightarrow$ columns span $\mathbb{R}^4$. In other words the columns of A span all $\mathbb{R}^4$ so $A\mathbf{x} = \mathbf{b} \quad \forall \mathbf{b} \in \mathbb{R}^4$
- In the false case, i.e. if some row has no pivot, there will be some $\mathbf{b}$ for which $A\mathbf{x} = \mathbf{b}$ has no solution.
- **17.** When you reduce it, you will find that A has 3 pivot positions (rows) and thus there is not a solution for each $\mathbf{b}$ in $\mathbb{R}^4$
- **19.** Not each vector can be written as a linear combination of matrix A meaning not every column spans $\mathbb{R}^4$

31. Let A be a 3×2 matrix. Explain why the equation $A\mathbf{x} = \mathbf{b}$ cannot be consistent $\forall \mathbf{b} \in \mathbb{R}^3$. Generalize your argument to the case of an arbitrary A with more rows than columns

– Draw a 3×2 matrix, more rows than columns.

$$A\mathbf{x} = \begin{bmatrix} * & * \\ * & * \\ * & * \end{bmatrix} \mathbf{b} = \begin{bmatrix} * \\ * \\ * \end{bmatrix}$$

- Conjecture: Proof by conjecture. Assume $Ax = b$ can be consistent $\forall b \in \mathbb{R}^3$. Then as we do RREF the last row will be all zero. Thus when we combine the augmented matrix $Ax = b$ we will see that b is a pivot for R^3 . Implying $\neg R$. Thus the system is inconsistent.
- **Definition:** A matrix cannot have more pivots than columns, so. A can have at most 2 pivots.
- **Better proof:**
 - * For any $m \cdot n$ matrix A with $m > n$ (more rows than columns), the maximum number of pivots is $n < m$, so there will **always** be at least one row without a pivot. Thus $Ax = b$ cannot be consistent $\forall b \in \mathbb{R}^m$

Homework 1.5

We use parametric vector form because $Ax = 0$ defines a subspace, and parametric vectors are the cleanest way to describe all vectors in that subspace using a small set of direction vectors.

- 7 & 11. Describe all solutions of $Ax = 0$ in parametric vector form, where A is row equivalent to the given matrix
- This is testing that you can solve with REF and REF, and factor out the free variables s.t. you write it in parametric vector form to represent how formulas look over the span
 - [Walkthrough](#)
 - The textbook solution doesn't write in parametric form, and they also reduce it fully to obtain the general solution $= 0$, before they write it in parametric form.
15. Describe and compare the solution sets of $x_1 + 5x_2 - 3x_3 = 0$ and $x_1 + 5x_2 - 3x_3 = -2$
- This question is testing that you know how to write a parametric solution with a nontrivial solution; or how I like to call them, particular solutions. Defined as:
 - * $\mathbf{p}$ = one specific solution, constants go within it
 - * $Ax = b$ is $Ap = b$
 - You express the equations trivial solution, and then non trivial solution as $p + sv_1 + tv_2$. Giving:

$$\begin{bmatrix} -2 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} -5 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

29. A is a 4×4 matrix with three pivot positions. Evaluate the cases:

1. Does $Ax = 0$ have a nontrivial solution?
2. Does the equation $Ax = b$ have at least one solution for all $b \in \mathbb{R}^4$?
 - False, By our definition, nontrivial solution $\iff A$ has a free variable. Since we have $4 \times 4 = A$ with only three pivots, it is trivial.
 - Since our system of equations A has a free variable, it is not the case that there are solutions for all b in $\mathbb{R}^4$. More formally

$$\exists x \in \mathbb{R}^4 \text{ s.t. } Ax = b$$

31. A is a 3×2 matrix with two pivot positions.

$$\begin{bmatrix} \square & * & * \\ * & \square & * \end{bmatrix}$$

- It is trivial because we have a pivot in every column
- $\exists b \in \mathbb{R}^3 Ax = b$. So to reiterate, there exists a solution, but not for all b .

Homework 1.7

3. Tests your understanding of viewing simple perfect square matrices to find dependence

$$\begin{bmatrix} 2 \\ -4 \end{bmatrix} \begin{bmatrix} -3 \\ 6 \end{bmatrix}$$

True, linearly dependent bc $-2u = v$

7. Determine if the columns of the matrix form a linearly independent set.

$$\begin{bmatrix} 1 & 4 & -3 & 0 \\ -2 & -7 & 5 & 1 \\ -4 & -5 & 7 & 5 \end{bmatrix}$$

You can row reduce to see that the last column will not have a pivot. Or you can observe that we have more columns than entries and assume that the last column will not contain a pivot

11. Find value(s) of h for which the vectors are linearly dependent

Placeholder equation I need to write

Problem asks you to simply do row operations to see what makes it a pivot, and then solve for h s.t. you make h the value that will make it a free vector, and thus it is nontrivial and linearly dependent.

Homework 1.8

3. T defined by $T(x) = A(x)$, find vector x whose image under T is $\mathbf{b}$, and determine whether $\mathbf{x}$ is unique.

- This question tests your ability to use notation with what we've done in the past.
- The steps are as follows.
 - (a) RREF
 - (b) General Solution, write as $\mathbf{x}$
 - (c) Determine uniqueness

7. Let A be a 6×5 matrix. What must $a \wedge b$ be in order to define $T : \mathbb{R}^a \rightarrow \mathbb{R}^b$ by $T(x) = A(x)$

- (a) This question is testing your understanding of how we think of A , x , and the process of creating a vector of Ax that has one vector, by the dot product rule.

11. Let $\mathbf{b} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$ and let A be the matrix in Exercise 9. Is $\mathbf{b}$ in the range of the linear transformation $x \mapsto Ax$? Why or why not.

- This question is just another wording for you to set up an augmented matrix and see if it's consistent.

27. Let u and v be linearly independent vectors in $\mathbb{R}^3$, and let P be the plane through u, v and 0 . The parametric equation of P is $x = su + tv$. Show that a linear transformation $T : \mathbb{R}^3 \mapsto \mathbb{R}^3$ maps P onto a plane through 0 , or onto a line through 0 , or onto just the origin in $\mathbb{R}^3$. What must be true about $T(u)$ and $T(v)$ in order for the image of the plane P to be a plane?

Given Information We are given that the linearly independent vectors u and v are in $\mathbb{R}^3$ and a plane which passes through the vectors u, v and 0 . The parametric equation of the plane is

$$\mathbf{x} = s\mathbf{u} + t\mathbf{v}$$

We have to show that a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ maps P onto a plane through 0 or onto a line through 0 , or just the origin in $\mathbb{R}^3$. Step-1: Since T is a linear transformation,

$$\begin{aligned} T(\mathbf{x}) &= T(s\mathbf{u} + t\mathbf{v}) \\ &= T(s\mathbf{u}) + T(t\mathbf{v}) \\ &= sT(\mathbf{u}) + tT(\mathbf{v}) \end{aligned}$$

Step-2: If $T(\mathbf{u})$ and $T(\mathbf{v})$ are linearly independent vectors in $\mathbb{R}^3$, then the parametric equation of a plane in $\mathbb{R}^3$ is:

$$T(\mathbf{x}) = sT(\mathbf{u}) + vT(\mathbf{v})$$

Given that the plane P passes through the vector 0 and T is a linear transformation, then $T(0) = 0$. So that, the plane $T(\mathbf{x}) = sT(\mathbf{u}) + vT(\mathbf{v})$ passed through the origin. So, T is a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^3$ that maps P onto a plane through 0 when $T(u)$ and $T(v)$ are linearly independent Step-3: In case, $T(\mathbf{u})$ and $T(\mathbf{v})$ are linearly dependent and both are nonzero, then one vector can be written as a linear combination of other two vectors.

$$T(\mathbf{u}) = cT(\mathbf{v})$$

A line in the direction of $T(v)$ is

$$T(\mathbf{x}) = scT(\mathbf{v}) + vT(\mathbf{v}) = (sc + v)T(\mathbf{v})$$

Since the line P passes through origin and T is a linear transformation, then $T(\mathbf{x}) = sT(\mathbf{u}) + vT(\mathbf{v})$ should pass through the origin. Therefore, there is a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that maps P onto a line passing through origin when the vectors $T(\mathbf{u})$ and $T(\mathbf{v})$ are linearly dependent and are nonzero. Step-4: If both the vectors are equal to 0 , then $T(x) = 0$. hence $x = 0$. Thus, the parametric equation represents only origin. Therefore, there is a linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that maps P onto origin when the vectors $T(\mathbf{u})$ and $T(\mathbf{v})$ are both zero

[Reference Image](#)

- The condition for the set of images of the plane P under the linear transformation T to be a plane through 0 is that $T(u)$ and $T(v)$ must be linearly independent. The condition for the set of images to be a line through 0 is that $T(u)$ and $T(v)$ must be parallel or equal to each other. The condition for the set of images to be just the origin in $\mathbb{R}^3$ is that $T(u)$ and $T(v)$ must be zero vector

32. Show that the transformation T defined by $T(x_1, x_2) = (4x_1 - 2x_2, 3|x_2|)$ is not linear

- The equation

$$4x_1 - 2x_2, 3|x_2|$$

is not linear because its an absolute value, and thus it wont span everything. **So lets note that**

||

and constants make it not linear

Homework 1.9

3. $\mathbb{R}^2 \rightarrow \mathbb{R}^2$, is a vertical shear transformation that maps $\mathbf{e}_1$ into $\mathbf{e}_1 - 3\mathbf{e}_2$, but leaves $\mathbf{e}_2$ unchanged. [Linear Transformation Matrix]

$$T(x) = [T(\mathbf{e}_1) \ T(\mathbf{e}_2)]x$$

- This question is testing your knowledge of the theorem that $T(x) = Ax$.
- Recall that shear transformation is:

$$\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$$

- Simply apply the change to $\mathbf{e}_1$ by the formula to get

$$\begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}$$

- Since we know k, we don't include it in our answer

7. $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ rotates points through $-\frac{3\pi}{4}$ radians (clockwise) and then reflects points through the horizontal x_1 axis. Hint: $T(e_1) = (-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}})$

- This question is asking you to plug in $\cos(x) \wedge \sin(x)$ where $x = -\frac{3\pi}{4}$ to do scale operation
- Then from there you reflect it.

44. Let $S : \mathbb{R}^p \rightarrow \mathbb{R}^n$ and $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be linear transformations. Show that the mapping $\mathbf{x} \mapsto T(S(\mathbf{x}))$ is a linear transformation from $\mathbb{R}^p \rightarrow \mathbb{R}^m$

The given statement "The mapping $\mathbf{x} \mapsto T(S(\mathbf{x}))$ is a linear transformation from $\mathbb{R}^p$ to $\mathbb{R}^m$ " is true. Because it satisfies the properties of linearity, as shown by the computation of $T(S(c\mathbf{u} + d\mathbf{v}))$. To show that the mapping $\mathbf{x} \mapsto T(S(\mathbf{x}))$ is a linear transformation from $\mathbb{R}^p$ to $\mathbb{R}^m$, we need to show that it satisfies the following two properties for all vectors $\mathbf{x}, \mathbf{y}$ in $\mathbb{R}^p$ and all scalars c, d : Additivity: $T(S(c\mathbf{x} + d\mathbf{y})) = cT(S(\mathbf{x})) + dT(S(\mathbf{y}))$ Homogeneity: $T(S(c\mathbf{x})) = cT(S(\mathbf{x}))$ Let's first show the additivity property: $T(S(c\mathbf{x} + d\mathbf{y})) = T(S(c\mathbf{x} + d\mathbf{y}))$ [Definition of vector addition] $= T(S(c\mathbf{x}) + S(d\mathbf{y}))$ [By linearity of S] $= T(S(c\mathbf{x})) + T(S(d\mathbf{y}))$ [By linearity of T] $= cT(S(\mathbf{x})) + dT(S(\mathbf{y}))$ [By linearity of S and T] Therefore, $T(S(\mathbf{x}))$ satisfies the additivity property. Now let's show the homogeneity property: $T(S(c\mathbf{x})) = T(S(c\mathbf{x}))$ [Definition of scalar multiplication] $= cT(S(\mathbf{x}))$ [By linearity of S and T] Therefore, $T(S(\mathbf{x}))$ satisfies the homogeneity property. Since $T(S(\mathbf{x}))$ satisfies both the additivity and homogeneity properties, it is a linear transformation from $\mathbb{R}^p$ to $\mathbb{R}^m$

1 Lecture 1.5 Comments from Professor

Ex. Say you have 50 vectors in $\mathbb{R}^{100}$. Do they create something like $\mathbb{R}^{50}$?

1. maybe?
2. How do we account for redundancy; i.e. calculating the redundant linear equations

Prof said that we can see in 2*2 matrix when something is linearly dependent, i.e. the two equations are multiples of each other.

1. But he cautioned saying that this is not the correct way to think about independence.
2. Because in A , we will have many spans of $\mathbb{R}^m$, and then we need to determine multiples and sums of equations that are dependent on each other

Prof also cautioned against the blind assumption that if we have a 5*5 matrix, and we are asked to find the span of R^3 , to not blindly assume that there are two pivots that are free. Because while it is a correct assumption, we don't understand why.

Thus the answer to "How do I pick which linear equations are redundant?" is attentively we use **RREF** to see **linear independence** by identifying **pivots** and understanding **triviality**.

Homework 2.1

3. Let

$$A = \begin{bmatrix} 4 & -1 \\ 5 & -2 \end{bmatrix}. \text{ Find } 3I_2 - A, (3I_2)A$$

Proofs to solve these questions:

Subtraction (Vice-versa for addition):

$$A - B := [a_{ij} - b_{ij}] \quad \forall \quad 1 \leq i \leq n, 1 \leq j \leq m \quad \wedge \quad n = m \in \mathbb{F} \rightarrow$$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \\ a_{11} & a_{12} & a_{13} \end{bmatrix} \quad B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{11} & b_{12} & b_{13} \\ b_{11} & b_{12} & b_{13} \end{bmatrix} \rightarrow$$

$$A - B = \begin{bmatrix} a_{11} - b_{11} & a_{12} - b_{12} & a_{13} - b_{13} \\ a_{11} - b_{11} & a_{12} - b_{12} & a_{13} - b_{13} \\ a_{11} - b_{11} & a_{12} - b_{12} & a_{13} - b_{13} \end{bmatrix}$$

Multiplication:

$$(cI_n)A_{ij} = \sum_k c\delta_{ik}A_{kj}$$

By Kronecker identity $(cI_n)A_{ij} = \sum_k c\delta_{ik}A_{kj} = A_{ij} \rightarrow cA_{ij}$

7. If a matrix A is 5×3 and the product AB is 5×7 , what is the size of B ?

- The definition of **defined** is being tested.
- For AB to be defined, it must satisfy the condition that $A^{m \times n} B^{n \times p}$.
- In other words, $AB \iff \text{rows} \cdot \text{columns}$
- Thus, the only logical answer is that the A^i matches with B^j . Where $j = \text{column}$ and $i = \text{row}$. Meaning the answer is $C^{3 \cdot 7}$

11. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$ and $D = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Find AD and DA . Explain how the columns or rows of A change when A is multiplied by D on the right or on the left. Find a 3×3 matrix B , not the identity matrix or the zero matrix, s.t. $AB = BA$

We use the proof for columns and rows Case j : $(AD)_{ij} = \sum_k A_{ik}D_{kj} \implies k = j$ term survives

Case i : $(DA)_{ij} = \sum_k A_{ik}D_{kj} \implies k = i$ term survives

In laymans terms, this means when I is to the right that we apply the pivot to all A_j variables.

and

When I is to the left we apply the pivot to the A_i variables.

For $AB = BA$ use the definition that:

- We can write B as λI .
- λ represents the scalar and I_n is the identity with n rank. Thus:

Let $A \in M_{n \times n}$, Let $B = \lambda I_n \wedge I \neq 0, 1 \rightarrow$

$$AB = A(\lambda I_n) = \lambda(I_n A) = \lambda A$$

$\wedge$

$$BA = (\lambda I_n)A = \lambda(I_n A) = \lambda A$$

Thus, $AB = BA$ ■

27. Suppose the third column of B is the sum of the first two columns. What can you say about the third column of AB ?

- If B has dependence, then it is not invertible, thus it follows that AB will also be dependent and singular because $\det(B) = 0 \implies \det(AB) = 0$
- We can prove that AB third column will be $A(b_1) + A(b_2)$ because B_3 third column is $b_3 = b_1 + b_2$.
- We can also state that there will be a lower dimension output.

The real idea is that linear transformations preserve linear relations.

Homework 2.2

3. Find inverse of the matrix $\begin{bmatrix} 8 & 3 \\ -7 & -3 \end{bmatrix}$

- Apply the algorithm:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \rightarrow \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \rightarrow ad - bc \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = A^{-1}$$

- Solution is

$$\begin{bmatrix} 1 & 1 \\ -\frac{7}{3} & -\frac{8}{3} \end{bmatrix}$$

7. Find inverse of the system of equations and then use it to solve the system

$$\begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix}$$

Helpful note:

$$Ax = B \text{ is } \begin{bmatrix} 8 & 3 \\ 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

- Convert A into A^{-1} . With the definition of $A^{-1}A = I$ we show that:

We show $x = A^{-1}B$

$$Ax = B$$

$$A^{-1}Ax = A^{-1}B$$

$$Ix = A^{-1}B$$

$$x = A^{-1}B$$

- Thus we multiply A^{-1} by B .
- We must recognize the dimensions our new expression will be in. I.e. $\mathbb{R}^{2 \times 2} \wedge \mathbb{R}^{2 \times 1}$ to see our product will be $\mathbb{R}^{2 \times 1}$.

$$\begin{bmatrix} -4 & -3 \\ 10 & 8 \end{bmatrix} \rightarrow \begin{bmatrix} -7 \\ 18 \end{bmatrix}$$

11. In order for a matrix B to be the inverse of A , both equations $AB = I$ and $BA = I$ must be true

- This is testing your knowledge of the theorem of inverse.
- If square matrix then $AB = I \wedge BA = I$

13. If A and B are $n \times n$ and invertible, then $A^{-1}B^{-1}$ is the inverse of AB .

- This statement would be true if it stated $A^{-1}B^{-1}$ is the inverse of BA , as that is the definition

21. Let A be an invertible $n \times n$ matrix, and let B be a $n \times p$ matrix. Show that the equation $AX = B$ has a unique solution $A^{-1}B$.

- We are being asked to prove $\exists$ a solution and if its $\exists$!
- To show existence, we check if dimensions let use define. In this case,
 $n \times n \wedge n \times p = n \times p$ ✓

$$\begin{aligned} AX &= \\ A(A^{-1}B) &= \\ AA^{-1}(B) &= \\ I(B) &= B \end{aligned}$$

- To check uniqueness we multiply A^{-1} on the leftside on each side of the expression.

$$\begin{aligned} AX &= B \\ (A^{-1})AX &= (A^{-1})B \\ (A^{-1}A)X &= (A^{-1})B \\ I(X) &= A^{-1}B \\ X &= A^{-1}B \end{aligned}$$

23. Let $AB = AC$, where B and C are $n \times p$ matrices and A is invertible. Prove $B = C$. Is it true when A is not invertible.

- We can show that $B = C$ by simply applying our formula $A^{-1}A = I$. This proves its unique as well.
- It is true that we can divide A from $AB = AC$, but cancellation is not valid because A is singular. Thus it is the case that $B = C$ is not possible, while we can have the condition that $AB = AC$. The counter example is:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} C = \begin{bmatrix} 0 & 2 \\ 0 & 0 \end{bmatrix}$$